UD–UAV offloading

Direct UD–LEO offloading

Local computing

UAV trajectory

State / control signaling

Edge server

Deadline-based task priority

Task {D, C, τ}

Priority ω = 1/τ

Shorter deadline, higher priority

LEO satellite with edge computing



600 km altitude

100 MHz shared, 50 GHz

Logical centralized decision maker

Offloading and bandwidth / CPU allocation

UAV trajectory control

UAV 1



10 MHz, 10 GHz

UAV 2



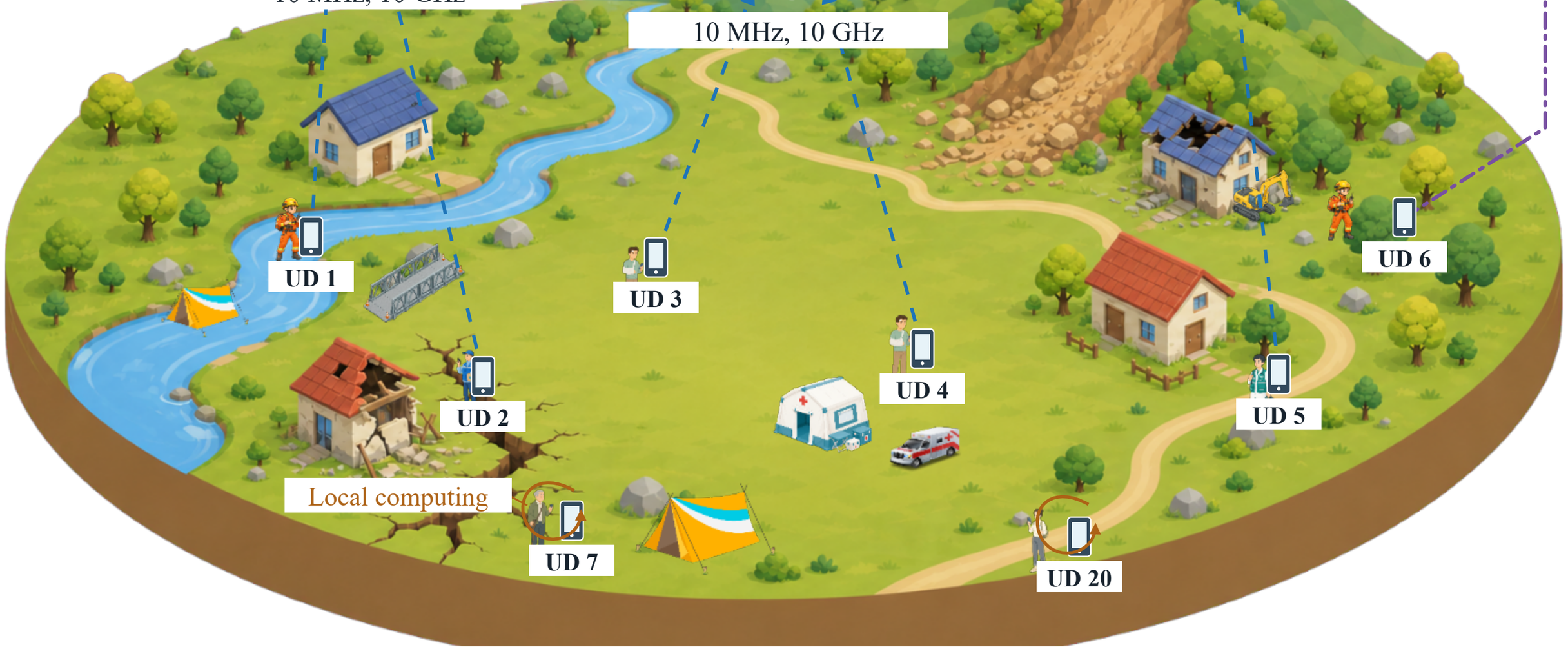
10 MHz, 10 GHz

UAV 3



10 MHz, 10 GHz

Direct UD–LEO offloading



Post-disaster rescue area: 20 mobile UDs with local computing

Each task executes at exactly one destination: local, one UAV, or the LEO satellite